

Wall-X LeRobot Media VisPruner 数据统计报告

Wall-X / VisPruner 实验整理

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数据来源

本报告由 `libero_media_40img_20vid_keep_0.5_results.json` 自动整理生成。实验使用数据集 `libero_all`，图片样本 40 个，视频样本 20 个，`keep_ratio` 为 0.5，设备为 `cuda`。所有耗时单位均为毫秒；差值为“裁剪配置 - 基线”，负数表示更快。

核心统计表

类型	样本数	平均差 (ms)	中位差 (ms)	最小差 (ms)	最大差 (ms)	标准差 (ms)	平均变化	更快数	更慢数
图片样本	40	-2.209	-1.655	-23.219	+13.870	6.173	-0.64%	27	13
视频样本	20	-11.199	-9.294	-45.976	-1.708	9.036	-2.61%	20	0

图片样本明细

序号	来源	裁剪前 token	裁剪后 token	token 变化	基线总耗时	裁剪总耗时	耗时差	耗时变化
1	idx=0	81	41	-49.38%	335.957	334.273	-1.684	-0.50%
2	idx=6968	81	41	-49.38%	344.540	333.626	-10.915	-3.17%
3	idx=13936	81	41	-49.38%	335.420	331.246	-4.174	-1.24%
4	idx=20905	81	41	-49.38%	335.347	335.975	+0.627	+0.19%
5	idx=27873	81	41	-49.38%	336.581	340.271	+3.690	+1.10%
6	idx=34842	81	41	-49.38%	333.330	330.037	-3.294	-0.99%
7	idx=41810	81	41	-49.38%	334.802	330.066	-4.735	-1.41%
8	idx=48779	81	41	-49.38%	333.800	331.694	-2.106	-0.63%
9	idx=55747	81	41	-49.38%	348.695	329.452	-19.243	-5.52%
10	idx=62716	81	41	-49.38%	341.737	329.653	-12.084	-3.54%
11	idx=69684	81	41	-49.38%	333.414	325.299	-8.115	-2.43%
12	idx=76653	81	41	-49.38%	338.473	337.100	-1.373	-0.41%
13	idx=83621	81	41	-49.38%	331.819	330.963	-0.856	-0.26%
14	idx=90590	81	41	-49.38%	335.818	331.364	-4.454	-1.33%
15	idx=97558	81	41	-49.38%	334.570	338.405	+3.835	+1.15%
16	idx=104527	81	41	-49.38%	331.151	333.752	+2.601	+0.79%
17	idx=111495	81	41	-49.38%	329.565	333.070	+3.506	+1.06%

序号	来源	裁剪前 token	裁剪后 token	token 变化	基线总耗时	裁剪总耗时	耗时差	耗时变化
18	idx=118464	81	41	-49.38%	329.736	333.826	+4.091	+1.24%
19	idx=125432	81	41	-49.38%	332.885	335.020	+2.136	+0.64%
20	idx=132401	81	41	-49.38%	332.585	335.300	+2.715	+0.82%
21	idx=139369	81	41	-49.38%	332.771	330.489	-2.282	-0.69%
22	idx=146338	81	41	-49.38%	334.584	331.801	-2.782	-0.83%
23	idx=153306	81	41	-49.38%	334.776	330.503	-4.273	-1.28%
24	idx=160275	81	41	-49.38%	333.480	331.853	-1.627	-0.49%
25	idx=167243	81	41	-49.38%	329.813	325.037	-4.775	-1.45%
26	idx=174212	81	41	-49.38%	332.115	330.493	-1.622	-0.49%
27	idx=181180	81	41	-49.38%	333.151	334.412	+1.261	+0.38%
28	idx=188149	81	41	-49.38%	328.362	327.436	-0.926	-0.28%
29	idx=195117	81	41	-49.38%	332.145	333.647	+1.502	+0.45%
30	idx=202086	81	41	-49.38%	334.638	348.507	+13.870	+4.14%
31	idx=209054	81	41	-49.38%	328.649	323.644	-5.005	-1.52%
32	idx=216023	81	41	-49.38%	329.399	331.900	+2.500	+0.76%
33	idx=222991	81	41	-49.38%	330.979	329.830	-1.149	-0.35%
34	idx=229960	81	41	-49.38%	329.964	328.968	-0.996	-0.30%
35	idx=236928	81	41	-49.38%	351.754	328.535	-23.219	-6.60%
36	idx=243897	81	41	-49.38%	335.717	332.901	-2.816	-0.84%
37	idx=250865	81	41	-49.38%	337.530	333.693	-3.837	-1.14%
38	idx=257834	81	41	-49.38%	335.675	333.600	-2.076	-0.62%
39	idx=264802	81	41	-49.38%	333.358	334.838	+1.479	+0.44%
40	idx=271771	81	41	-49.38%	334.370	332.618	-1.752	-0.52%

视频样本明细

序号	来源	裁剪前 token	裁剪后 token	token 变化	基线总耗时	裁剪配置总耗时	耗时差	耗时变化
1	episode_000000.mp4	324	324	0.00%	416.131	410.703	-5.428	-1.30%
2	episode_000001.mp4	324	324	0.00%	419.307	403.206	-16.101	-3.84%
3	episode_000002.mp4	324	324	0.00%	427.495	412.213	-15.282	-3.57%
4	episode_000003.mp4	324	324	0.00%	428.389	415.307	-13.082	-3.05%
5	episode_000004.mp4	324	324	0.00%	418.656	412.776	-5.880	-1.40%
6	episode_000005.mp4	324	324	0.00%	434.178	417.201	-16.977	-3.91%
7	episode_000006.mp4	324	324	0.00%	417.537	415.829	-1.708	-0.41%
8	episode_000007.mp4	324	324	0.00%	422.269	414.401	-7.868	-1.86%
9	episode_000008.mp4	324	324	0.00%	426.706	412.340	-14.366	-3.37%
10	episode_000009.mp4	324	324	0.00%	455.490	409.514	-45.976	-10.09%
11	episode_000010.mp4	324	324	0.00%	417.260	408.041	-9.219	-2.21%
12	episode_000011.mp4	324	324	0.00%	415.233	410.816	-4.416	-1.06%
13	episode_000012.mp4	324	324	0.00%	414.014	410.878	-3.136	-0.76%
14	episode_000013.mp4	324	324	0.00%	415.582	408.195	-7.387	-1.78%
15	episode_000014.mp4	324	324	0.00%	418.712	409.343	-9.369	-2.24%
16	episode_000015.mp4	324	324	0.00%	423.647	410.613	-13.034	-3.08%
17	episode_000016.mp4	324	324	0.00%	421.311	412.975	-8.336	-1.98%
18	episode_000017.mp4	324	324	0.00%	426.928	420.819	-6.109	-1.43%
19	episode_000018.mp4	324	324	0.00%	421.265	411.610	-9.655	-2.29%
20	episode_000019.mp4	324	324	0.00%	422.653	412.001	-10.653	-2.52%

图片阶段耗时完整表

阶段	基线 (ms)	裁剪 (ms)	差值 (ms)	变化
vision_image_encode	26.279	0.000	-26.279	-100.00%

阶段	基线 (ms)	裁剪 (ms)	差值 (ms)	变化
vision_image_encode_score	0.000	24.522	+24.522	0.00%
total_time	334.586	332.378	-2.209	-0.66%
ode_integration	276.006	275.024	-0.982	-0.36%
ode_transformer_total	269.724	268.754	-0.970	-0.36%
vispruner_total	0.000	0.804	+0.804	0.00%
position_encoding	0.620	0.108	-0.512	-82.58%
pruning_position_ids_prepare	0.000	0.504	+0.504	0.00%
position_ids_rope	0.491	0.000	-0.491	-100.00%
embed_processing	26.768	26.363	-0.405	-1.51%
image_path_total	26.370	25.993	-0.376	-1.43%
vision_image_forward	26.395	26.020	-0.375	-1.42%
prefill_transformer	29.200	28.898	-0.302	-1.03%
prefetch_forward	29.392	29.092	-0.300	-1.02%
vispruner_build_keep_mask	0.000	0.223	+0.223	0.00%
vispruner_apply_keep_to_sequences	0.000	0.188	+0.188	0.00%
vispruner_topk_select	0.000	0.127	+0.127	0.00%
vispruner_rope_deltas	0.000	0.123	+0.123	0.00%
external_prepare_batch_ms	3.661	3.573	-0.088	-2.42%
vispruner_pad_pruned_batch	0.000	0.072	+0.072	0.00%
vispruner_image_lengths	0.000	0.057	+0.057	0.00%
vispruner_gather_image_embeds	0.000	0.039	+0.039	0.00%
vispruner_score_prepare	0.000	0.026	+0.026	0.00%
scatter_image_embeds	0.116	0.096	-0.020	-16.92%
ode_prepare_inputs	0.626	0.619	-0.007	-1.14%
action_initialization	0.428	0.424	-0.004	-0.97%
scatter_proprioception	0.123	0.119	-0.004	-3.34%
kv_cache_trim	0.810	0.814	+0.003	+0.43%

阶段	基线 (ms)	裁剪 (ms)	差值 (ms)	变化
cache_preprocessing	1.238	1.234	-0.003	-0.27%
postfix_mask_build	0.122	0.119	-0.003	-2.34%
image_cast	0.043	0.040	-0.003	-6.27%
action_init_embed	0.258	0.255	-0.003	-1.03%
prefix_length_resolve	0.067	0.065	-0.002	-2.63%
embed_tokens	0.039	0.037	-0.001	-3.81%
action_init_noise	0.045	0.043	-0.001	-2.58%
moe_indices	0.086	0.087	+0.001	+1.08%
postfix_moe_indices	0.099	0.098	-0.001	-0.93%
scatter_action_init	0.056	0.055	-0.001	-1.61%
ode_action_embed_total	2.637	2.638	+0.000	+0.01%
attention_mask_to_device	0.006	0.006	-0.000	-3.27%
postprocessing	0.009	0.008	-0.000	-2.41%
postfix_slice	0.054	0.054	-0.000	-0.34%
prefill_action_head	0.140	0.140	+0.000	+0.11%
ode_action_head_total	0.939	0.939	+0.000	+0.02%

视频阶段耗时完整表

阶段	基线 (ms)	裁剪配置 (ms)	差值 (ms)	变化
vision_video_forward	107.405	91.231	-16.174	-15.06%
embed_processing	107.880	91.750	-16.131	-14.95%
total_time	423.138	411.939	-11.199	-2.65%
ode_integration	282.757	286.635	+3.878	+1.37%
ode_transformer_total	276.246	279.280	+3.034	+1.10%

阶段	基线 (ms)	裁剪配置 (ms)	差值 (ms)	变化
prefetch_forward	29.898	30.767	+0.869	+2.91%
prefill_transformer	29.696	30.540	+0.844	+2.84%
external_prepare_batch_ms	19.597	19.141	-0.456	-2.33%
ode_action_embed_total	2.723	3.074	+0.352	+12.91%
ode_action_head_total	0.975	1.180	+0.205	+21.05%
cache_preprocessing	1.267	1.339	+0.073	+5.73%
position_encoding	0.736	0.806	+0.070	+9.51%
position_ids_rope	0.598	0.658	+0.060	+9.99%
ode_prepare_inputs	0.638	0.687	+0.048	+7.60%
action_initialization	0.461	0.501	+0.040	+8.70%
action_init_embed	0.280	0.311	+0.031	+11.21%
kv_cache_trim	0.812	0.842	+0.030	+3.69%
prefill_action_head	0.153	0.176	+0.023	+15.11%
scatter_proprioception	0.141	0.159	+0.019	+13.35%
postfix_moe_indices	0.105	0.124	+0.018	+17.45%
postfix_mask_build	0.133	0.146	+0.013	+10.08%
moe_indices	0.093	0.101	+0.009	+9.47%
prefix_length_resolve	0.078	0.085	+0.007	+8.36%
scatter_video_embeds	0.062	0.067	+0.005	+8.78%
embed_tokens	0.048	0.052	+0.004	+8.09%
action_init_noise	0.049	0.053	+0.003	+6.85%
postfix_slice	0.055	0.057	+0.003	+4.96%
scatter_action_init	0.058	0.060	+0.002	+3.24%
attention_mask_to_device	0.007	0.007	+0.000	+2.13%
postprocessing	0.009	0.009	+0.000	+0.16%

读表说明

表中的“更快数”统计的是总耗时差小于 0 的样本数量。阶段耗时表按差值绝对值从大到小排列，方便先查看变化最明显的模块。对于基线中不存在而裁剪配置中新增的阶段，例如 `vispruner_total`，变化百分比仅作新增开销参考，不应与普通加速比例直接等同。